A Graph-Benders Decomposition for Integrated Electric Trucks Routing, Cargo Loading and V2G Scheduling

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Abstract—The tightening of global environmental regulations is accelerating the adoption of Electric Trucks (ETs) in logistics; however, the integrated optimization of their routing, charging schedules, and cargo loading presents a formidable challenge. This complex coupling inherently introduces a Generalized Multiple Knapsack Problem and results in a Mixed-Integer Linear Programming model that is typically NP-hard and computationally demanding for standard solvers. To address this, we propose a novel Graph-Benders algorithm based on an exact algorithmic framework that strategically decomposes the problem into an ET routing master problem and a Vehicle-to-Grid (V2G) scheduling subproblem, reformulated within a graph-based modeling approach. This structure ensures all delivery tasks are fulfilled while tightening the feasible region for grid coordination. By synergizing feasibility and optimality cuts from the linearly relaxed subproblem with no-good cuts to address integer infeasibility, the method efficiently eliminates infeasible solutions,yielding a high-quality near-optimal solution, which is at least as good as the optimal V2G schedule for the best feasible routes.Through experiments in various geographic scenarios and split-delivery tasks, the algorithm demonstrates its versatility and achieves a breakthrough in efficiency without sacrificing solution quality compared to Gurobi. In large-scale tests, for instances with 19,585 decision variables, the proposed Graph-Benders decomposition yields a superior, high-quality near-optimal solution in merely 28 seconds, outperforming Gurobi's 10,000-second time-limit result. Notably, even at a scale of 127,489 variables, the algorithm yields high-quality near-optimal solutions in only 130 seconds.Consequently, the proposed algorithm effectively tackles the strong coupling of large-scale ET routing and V2G coordination, offering a powerful tool for practical low-carbon logistics deployment.

Index Terms—Electric Trucks; Vehicle Routing; Vehicle-to-Grid (V2G); Benders Decomposition; Graph-Based Optimization; Mixed-Integer Programming; Large-Scale Optimization

Nomenclature

Sets

$\mathcal{A}$	Set of simplified paths between nodes
$\mathcal{A}'$	Set of feasible roads between nodes
$\mathcal{N}$	Node set (including depots, destinations, customer nodes)
$\mathcal{N}'$	Node set (including depots, destinations, customer nodes, intersections)
$\mathcal{R}$	Set of customer nodes
Z	Set of charging/discharging time options, where $Z_o > 0$ denotes charging duration and $Z_o < 0$ denotes discharging duration.

Parameters

$E_k^{\max}$,$w_k^{\max}$	Maximum battery and load capacity of electric truck k
E_N^{set}	Electric truck energy setting at the terminal node
$G'(N', A')$	Actual transportation network graph
$G(N, A)$	Simplified transportation network graph
$P_{ch/dis,i}$,p_i	Charging/discharging power and price of the charging pile at node i
T_i^{req},Q_i,	Latest service time and total cargo weight at node i
Z_o	The o-th option for the charging/discharging time at the charging pile node
d_{ij},t_{ij},e_{ij},c_{ij}	Shortest distance, Travel time ,Energy consumption and Travel cost for electric trucks from nodes i to j
$SOC_{\max/\min}$	Maximum and Minimum state of charge for battery health

Decision Variables

c_i^k,τ_i^k	Continuous variable, representing charging/discharging amount/time of electric truck k at node i
q_i^k,b_i^k,t_i^k	Continuous variable, representing cargo load weight, remaining battery level and arrival time of electric truck k at node i
x_{ij}^k,$y_{i,o}^k$	Binary variable indicating whether electric truck k travels on path (i, j) and whether electric truck k selects charging/discharging option o at node i

Benders Algorithm

δ_k	Intercept of the unbounded ray
λ_{ij}^k	Dual values corresponding to constraints
$\overline{\Phi}$ $\underline{\Phi}$	Upper and Lower bound of the entire model
$\overline{x}_{ij}^k$	Solution generated by the master problem
$\varphi(x_{ij}^k)$	Auxiliary variable in master problem representing lower bound of subproblem

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K_{inf}	Set of electric trucks for which the subproblem is relaxed-feasible but integer-infeasible
S_k	Set of path edges selected by electric truck k in the current iteration, $S_k = \{(i,j) \in \mathcal{A} \mid x_{ij}^k = 1\}$
u_k	Dual unbounded ray of the subproblem

I. Introduction

Recently, International Energy Agency(IEA)'s Global EV Outlook 2025 underscores the rising pivotal role of electric trucks(ETs) in global transport decarbonization [1]. In 2024, global ET sales surged by nearly 80% year-on-year to exceed 90,000 units, with China accounting for over 80% of this growth—driven by scrappage schemes, falling battery costs, and tighter emission standards. Beyond sales momentum, e-trucks are gaining traction as their total cost of ownership (TCO) is projected to match diesel trucks in China and the EU by 2030 [2]. Despite lingering challenges like upfront costs and infrastructure gaps [3] , their expanding adoption highlights their irreplaceable role in shifting freight toward low-carbon systems, linking emissions reduction to long-term energy security [4].

ET routing's core challenge has shifted from simple range anxiety to tripartite optimization of routing, charging, and cargo loading. Key issues:limited battery energy density [5] requiring frequent charging and raising operational costs; long DC fast-charging times (40–60 minutes) conflicting with delivery time windows and forcing energy-delivery trade-offs; knapsack-like cargo loading from multi-node pickup/delivery complicating route sequencing and capacity allocation. Additionally, vehicle-to-grid (V2G) integration is hindered by conflicting scheduling objectives and lack of coordinated multi-vehicle algorithms, limiting ETs' economic/environmental potential. System-level co-optimization of routing, charging, and cargo loading has therefore become urgent for widespread ET adoption [6], [7].

Given practical challenges, research on spatiotemporal routing optimization and V2G-coordinated scheduling of ETs yields four key benefits:First, it redefines electric trucks as mobile storage units [8] rather than mere transport tools, providing a deeper economic rationale for their large-scale adoption from a strategic perspective. Second, by targeting the "idle windows" [5] inherent in the transport process—such as mandatory driver breaks and loading queues—the integration of these periods with V2G operations enables energy value-addition without imposing additional downtime. Furthermore, it emphasizes conducting V2G arbitrage without delaying any delivery tasks [9], thereby maximizing the value of stationary time while ensuring the reliability of logistics services. Finally, energy arbitrage revenue [10] offsets idle-time opportunity costs, accelerating TCO parity [2] while supporting grid stability via peak shaving.

The vehicle routing problem (VRP) underpins logistics scheduling [10], but deep ET-V2G integration exponentially increases complexity. Beyond conventional constraints (node sequences, load capacities), the model must incorporate EV-specific constraints: battery limits, spatial heterogeneity of charging infrastructure (variable charging power, geographical accessibility), and V2G temporal costs.Moreover, as electric logistics scenarios diversify and operational constraints grow more complex, the basic electric vehicle routing problem(EVRP) model fails to fully capture practical needs, leading to the emergence of application-oriented extensions. EVRP with Time Windows [11], which incorporates customer-specific time windows into the basic model, requiring services to be completed within designated intervals. its objective is to minimize total costs (travel, charging, time-window penalties), suitable for time-sensitive sectors (e.g., perishables, pharmaceuticals). EVRP with V2G [12] incorporates bidirectional energy exchange, allowing EVs to feed electricity back into the power system to coordinate logistics and energy demand. This synergy reduces operational costs and generates revenue, particularly for long-distance transport using V2G-enabled fleets [2]. EVRP with Split Delivery [13] relaxes the single-vehicle-per-customer constraint, allowing multiple trucks to serve one customer in separate trips, improving vehicle utilization and reducing total travel distance—suitable for high-demand scenarios. While key EVRP extensions—including time windows, V2G coordination, and split deliveries with inherent knapsack constraints—have been individually studied, their integrated optimization remains largely unexplored. This multi-faceted coupling creates a high-dimensional combinatorial space that grows exponentially, severely challenging commercial solvers.

The tight coupling of cargo, power, and coordination constraints makes the resulting EVRP strongly NP-hard. This complexity has established heuristic and metaheuristic methods as the dominant solution paradigm [9], which can be categorized into three groups: neighborhood search frameworks, including hybrid strategies like variable neighborhood search with tabu search [14]; swarm intelligence algorithms, such as improved genetic algorithms [15], particle swarm optimization [16], and hybrid architectures combining ant colony optimization with artificial bee colony [17]; and emerging data-driven methods like deep reinforcement learning [18] or learning-based graph feature extraction [19].

While these metaheuristic methods enhance high-dimensional search capabilities through problem-oriented neighborhood structures and adaptive operators, they lack scalability and struggle to balance solution speed with precision. Similarly, two-stage iterative approaches [20], which sequentially optimize routing and charging plans, suffer from similar limitations in scalability and solution accuracy. In contrast, exact methods such as Benders decomposition [21] guarantee optimality for small-scale instances; however, they are often hindered by slow convergence when applied to EVRPs—particularly those involving knapsack-like cargo loading constraints—due to an inability to fully leverage problem-specific characteristics.

In summary, despite these advancements, two critical research gaps remain unaddressed. First, existing literature largely treats EVRP variants—such as Time Windows, V2G coordination, or Split Delivery—in isolation, lacking a comprehensive framework to manage the tight tripartite coupling of spatiotemporal routing, knapsack-like cargo loading, and energy scheduling. This fragmentation limits the realization of system-level cost efficiencies and the full potential of grid interaction. Second, the resulting high-dimensional combinatorial complexity creates a severe computational bottleneck. As the problem scale increases, the efficiency of commercial solvers deteriorates rapidly, while standard exact decomposition methods often suffer from slow convergence due to weak cuts generated from the strongly coupled mixed-integer subproblems. These challenges necessitate a specialized optimization approach capable of balancing computational tractability with high-quality decision-making.

Recently, Zavala et al. proposed a graph-based modeling framework and its application within hierarchical Benders decomposition [22], [23], providing new insights into addressing strongly coupled decision-making challenges. Inspired by this approach, this paper develops an integrated graph-based model to concurrently capture the spatiotemporal routing, cargo loading, and V2G coordination of ETs. To achieve efficient resolution, we introduce a novel Graph Benders algorithm ,based on an exact logic-based Benders approach and tailored to this structure. The main contributions of this paper are summarized as follows:

- To address the operational requirements of coordinated transportation by multiple ETs, this paper establishes an integrated large-scale EVRP model that combines spatio-temporal routing with V2G coordination while accounting for cargo allocation. Using the Big-M method, the model reformulates the coupled time-route bilinear constraints (products of binary and continuous variables) into linear form, thereby enabling efficient computation while maintaining modeling accuracy.
- This paper presents the first introduction of graph-based modeling theory to the ET domain, establishing an integrated graph-theoretic framework that unifies multi-vehicle spatio-temporal routing optimization with V2G scheduling decisions. The work addresses a critical gap in existing EVRP research by providing a unified multi-level graph representation across temporal and spatial dimensions.
- Building upon the decomposable EVRP graph model for ETs , this study exploits the model's structural characteristic—specifically, that the global optimal solution aligns closely with the near-optimal solution of the optimal V2G dispatch under optimal routing. On this basis, we design an improved Benders decomposition algorithm. This algorithm integrates feasibility and optimality cuts derived from the linear relaxation of the subproblem, along with 'no-good cuts' to address integer infeasibility, while utilizing the variation rate of the global upper bound as a dynamic adjustment strategy. This hybrid mechanism significantly enhances solution efficiency for large-scale problems characterized by high-dimensional decision spaces, while ensuring the acquisition of high-quality approximate solutions.

II. System Formulation and Preprocessing

A. System Components and Graph Architecture

The proposed framework for coordinated spatio-temporal routing and V2G scheduling of ETs maintains a structure consistent with conventional scheduling models, comprising three core components: transportation network data, service node attributes, and ETs information. The model aims to minimize total transportation cost while completing all delivery tasks under constraints including time windows, cargo loading fulfillment, and battery capacity. Distinct from existing approaches, it incorporates a V2G coordination strategy that exploits spatial electricity price variations to optimize energy management. Specifically, trucks charge when passing through nodes with lower electricity prices and discharge back to the grid via V2G during high-price periods. This strategy not only improves power resource utilization but also effectively reduces overall operational costs.

(1) Transportation Network
The transportation network is modeled as an undirected graph $G'(N', A')$ (Fig. 1). The node set $N' = \{n_1, \ldots, n_n\}$ comprises critical points (depots, destinations, service nodes, intersections, etc.), where n_1 and n_n denote the origin and final destination. Each edge $(i, j) \in A'$ represents a road segment characterized by distance $d_{i,j}$ and travel time $t_{i,j}$.

(2) Service Nodes
A subset $R \subset N'$ (orange nodes in Fig. 1) denotes logistics service nodes. For each node $i \in R$, the characteristics are defined by the latest service request time T_i^{req}, cargo weight Q_i, charging power $P_{\text{ch},i}$, discharging power $P_{\text{dis},i}$, and the electricity price p_i. It is assumed that both charging and discharging processes are linear, where $P_{\text{ch},i}$ and $P_{\text{dis},i}$ can be distinct.

(3) Electric Trucks
Let K denote the set of ETs. For each vehicle $k \in K$, characteristics are defined by the max battery capacity $E_k^{\max}$ (operational range $[S_{k,\min}, S_{k,\max}]$), max load capacity $W_k^{\max}$, and unit distance cost u_k. The vehicles in K are indexed in non-decreasing order of $W_k^{\max}$.

Meanwhile, this work discretizes the charging and discharging time. Following common charging facility power configurations and user behavior patterns, the charging/discharging duration is divided into minute-level intervals (e.g., 5 minutes), generating multiple discrete power levels.

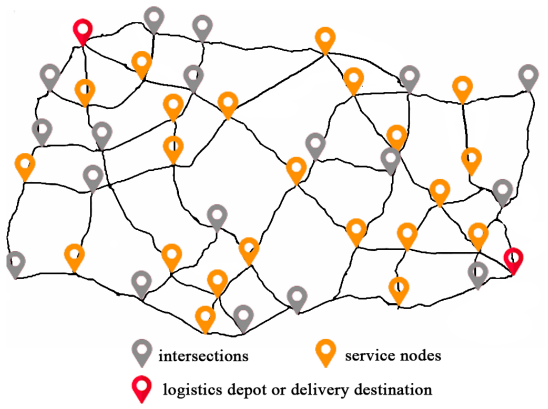


Fig. 1: Logistics Network Node Distribution Map

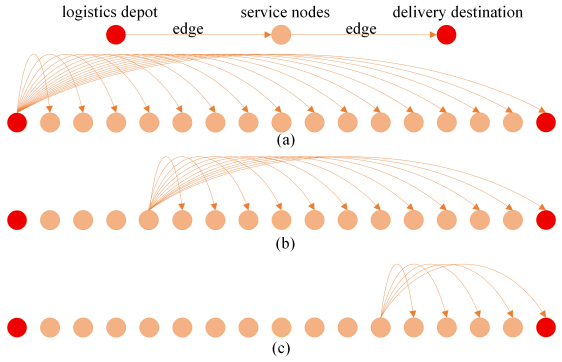


Fig. 2: Simplified Logistics Network :(a) routes from the origin depot, (b) and (c) connections among service request nodes

B. Preprocessing

To streamline the graph construction, this study employs a preprocessing technique [24] to transform the original physical graph $G'(N', A')$ into a reduced graph $G(N, A)$. The node set N comprises the origin depot, service nodes, and the destination depot (i.e., $N = \{n_1, n_n\} \cup R$).

Graph Reduction via Dijkstra's Algorithm.In the original graph G', nodes represent actual geographical locations, as illustrated in Fig. 1: red nodes denote the origin and destination depots for ETs, orange nodes represent service request nodes, and gray nodes indicate intersections or non-service nodes. To enhance computational efficiency and abstract the core logistics relationships, we apply Dijkstra's algorithm to G' to compute the shortest paths between all core nodes (red and orange nodes).

Topological Constraint through DAG Formulation. Subsequently, based on spatially determined node indices, we introduce a unidirectional path constraint that strictly enforces flow from lower- to higher-indexed nodes. This constraint effectively precludes reverse flows and cycles, structurally ensuring the network operates as a Directed Acyclic Graph . The resulting Simplified Logistics Network is shown in Fig. 2.

This strategy not only significantly prunes the search space by eliminating invalid branches but also forces the adjacency matrix into a strict upper triangular form, thereby transforming complex graph-theoretic searches into efficient linear algebraic operations. This transformation leads to a sharp decrease in the number of binary decision variables $x_{i,j}^k$. Consequently, the subsequent modeling and analysis will be based on the simplified graph $G(N, A)$ instead of the original graph $G'(N', A')$.

III. Spatio-Temporal EVRT and V2G Coordination Model for ETs

A. Model Formulation

$$\begin{aligned}\min W_1 \sum_{k=1}^{K} \sum_{(i,j)\in A} c_{ij} x_{ij}^k + W_2 \sum_{k=1}^{K} \sum_{i=1}^{N} p_i c_i^k + \\ W_3 \sum_{k=1}^{K} t_N^k + W_4 \sum_{k=1}^{K} \left(b_N^k - E_N^{\text{set}} \right)\end{aligned} \tag{1}$$

Objective: Equation (1) minimizes the total operational cost under fixed freight tasks, comprising four components: travel cost weighted by W_1 with c_{ij} denoting the per-kilometer cost; charging/discharging cost weighted by W_2, where p_i is the electricity price and c_i^k the energy exchange at node i; time cost weighted by W_3 based on arrival time t_N^k; and terminal residual energy deviation cost weighted by W_4,computed against the difference between actual battery level b_N^k and expected level E_N^{set}.

$$\sum_{j>i} x_{ij}^k - \sum_{j<i} x_{ji}^k = \begin{cases} 1, & \text{if } i \text{ is origin} \\ -1, & \text{if } i \text{ is destination} \\ 0, & \text{otherwise} \end{cases} \quad \forall k, i \tag{2}$$

Route Feasibility:To ensure the integrity and continuity of each truck k's route within the transportation network, the flow conservation constraint is introduced, as shown in Equation (2).In other words,this constraint fundamentally prevents path discontinuities, loops, or incorrect origin/destination assignments, establishing a reliable routing foundation for subsequent logistics scheduling and energy management.

$$\sum_{k=1}^{K} q_i^k = Q_i \quad \forall i \in N \tag{3}$$

$$q_i^k \le Q_i \sum_{j} x_{ij}^k \quad \forall i, k \tag{4}$$

$$\sum_{i=1}^{N} q_i^k \le W_k^{max} \quad \forall k \tag{5}$$

Cargo loading: To ensure all service node demands are fulfilled and no ET exceeds its load capacity during transportation, the cargo loading constraints shown in Equations (3)–(5) are introduced. Specifically:Equation (3) enforces strict global demand satisfaction, requiring the total fleet pickup to exactly match the demand ; Equation (4) establishes a logical coupling between routing and loading variables; it ensures cargo is loaded only conditional on a physical visit (i.e., forcing $q_i^k = 0$

if the vehicle does not visit); Equation (5) enforces the maximum cargo loading capacity for each ET k. In summary, these constraints integrate a Generalized Multiple Knapsack Problem into vehicle routing, ensuring all tasks remain within load capacities while satisfying demand and physical logic, thereby establishing a mathematical foundation for practical logistics schemes.

$$\sum_{i=1}^{N} q_i^k \le \sum_{i=1}^{N} q_i^{k+1}, \quad k, k+1 \in K \tag{6}$$

Additionally, to eliminate symmetry in homogeneous fleets, we enforce a non-decreasing cargo load order between adjacent vehicles k and $k+1$. This valid inequality effectively prunes the search space.

$$\sum_{o=1}^{O} y_{i,o}^k = \sum_{j \in N} x_{ij}^k \quad \forall i \ne 1, \forall k \tag{7}$$

$$c_i^k = P_{\mathrm{ch},i} \sum_{o:Z_o>0} y_{i,o}^k Z_o + P_{\mathrm{dis},i} \sum_{o:Z_o<0} y_{i,o}^k Z_o \quad \forall i,k \tag{8}$$

$$\tau_i^k = \sum_{o=1}^{O} y_{i,o}^k |Z_o| \quad \forall i \in N \tag{9}$$

Constrained Charging: Meanwhile, to describe the coupling relationship between charging/discharging decisions and vehicle routing at service nodes, this paper introduces the constraints shown in Equations (7)–(9). Specifically, Equation (7) represents the routing-charging/discharging coupling constraint, ensuring that only ET k visiting a node can select a charging/discharging mode there, with exactly one mode selection permitted. Equation (8) then determines the charging/discharging quantity c_i^k based on the selected mode $y_{i,o}^k$, where positive values indicate charging and negative values indicate discharging.Equation (9) precisely calculates the required charging/discharging duration according to the mode selection $y_{i,o}^k$, thus providing comprehensive temporal constraints for the model.

$$S_{\min} < b_i^k < S_{\max} \quad \forall i,k \tag{10}$$

$$S_{\min} < b_i^k + c_i^k < S_{\max} \quad \forall i,k \tag{11}$$

Equations (10) and (11) establish a dynamic constraint mechanism for battery capacity management. These constraints ensure that the battery capacity of ET k remains within the healthy operating range throughout the transportation process. By enabling control over charging and discharging states, this approach effectively mitigates battery degradation at the operational level, thereby extending the service life of ET batteries.

$$b_j^k = \left(b_i^k + c_i^k - e_{ij}\right) x_{ij}^k + \left(1 - x_{ij}^k\right) S_{\min}, \quad \forall (i,j) \in A, \forall k \tag{12}$$

$$t_j^k \ge \left(t_i^k + t_{ij} + \tau_i^k\right) x_{ij}^k, \quad \forall (i,j), k \tag{13}$$

$$t_i^k \le \sum_j x_{ij}^k T_i^{\mathrm{req}}, \quad \forall i,k \tag{14}$$

Finally, Equations (12)–(14) are introduced to maintain both temporal continuity and battery capacity continuity during ET transportation, while strictly satisfying the time window constraints at all service nodes. Specifically, Equations (12) and (13), formulated as bilinear constraints, ensure the continuity of energy and time states respectively: if ET k does not traverse a particular path, the battery capacity at the corresponding node is set to $S_{\min}$ and the time variable is set to zero. Equation (14) enforces a time window constraint, ensuring a truck's arrival at a node precedes its latest service time.

B. Linearization

To address the computational complexity and low solution efficiency caused by the nonlinear terms introduced in Equations (12) and (13), this paper employs the Big-M method for model linearization. Through appropriate relaxation techniques, the constraints are reformulated as follows:

$$b_j^k - b_i^k - c_i^k + e_{ij} \ge -E_k^{\max}(1 - x_{ij}^k) \quad \forall (i,j) \in A, \forall k \tag{15}$$

$$b_j^k - b_i^k - c_i^k + e_{ij} \le E_k^{\max}(1 - x_{ij}^k) \quad \forall (i,j) \in A, \forall k \tag{16}$$

$$b_j^k \le S_{\min}\left(1 - \sum_{i=1}^{N} x_{ij}^k\right) + S_{\max}\left(\sum_{i=1}^{N} x_{ij}^k\right) \quad \forall j \in N, \forall k \tag{17}$$

$$t_j^k \ge t_i^k + t_{ij} x_{ij}^k + \tau_i^k - T_N^{\mathrm{req}}(1 - x_{ij}^k) \quad \forall (i,j), k \tag{18}$$

Specifically, Equations (15) and (16) ensure that if ET k traverses path (i,j), an energy conservation relationship is maintained between the state of charge b_i^k at node i and b_j^k at node j. If the path is not selected, Equation (17) forces the state of charge at node j to be $S_{\min}$, thereby eliminating variable randomness and simplifying computation and data screening. Equation (18) represents the linearized form of Equation (13), guaranteeing temporal continuity.

It is worth noting that all constraints involving the Big-M method in this paper, such as Equations (14), (15), and (17), utilize tight physical upper and lower bounds (e.g., $E_k^{\max}$ and T_N^{req}) rather than arbitrarily large values. This approach is deliberately implemented to prevent the model formulation from becoming overly loose, thereby significantly improving solution efficiency.

In summary, through the aforementioned modeling and transformation steps, the problem is ultimately formulated as a mixed-integer linear programming (MILP) model. Although such problems can be solved using commercial solvers such as Gurobi and Cplex, the substantial scale of the model results in extended computation times and high computational resource requirements.

IV. Graph-Benders Decomposition Algorithm

A. Model Analysis and Benders Decomposition

Benders decomposition addresses complex optimization problems by decoupling them into a master problem and one or more subproblems. The master problem provides a lower bound solution, while the subproblems generate cuts to iteratively refine the solution via bound tightening until convergence is achieved .However, due to the inherent structure of this decomposition framework, achieving

rapid convergence for large-scale mathematical models remains challenging . The limited strength of the generated cutting planes often leads to slow convergence, even when enhanced with techniques such as Lagrangian relaxation [25] or regularization [26]. Nevertheless, the algorithm excels at obtaining high-quality near-optimal solutions quickly, as the cut-generation mechanism effectively eliminates solutions that are optimal for the master problem but infeasible for the subproblem early in the iterative process.

In the spatio-temporal EVRP and V2G coordination model for ETs, transportation costs dominate total operational expenses. The minimum-cost feasible path not only minimizes travel distance but also inherently provides sufficient time flexibility for en-route charging and discharging, thereby facilitating further reductions in second-stage V2G costs. In essence, the optimization direction of the Master Problem aligns closely with both the Subproblem's solution trajectory and the overall global optimization objective. Leveraging this structural characteristic, the optimal V2G schedules for the best feasible routes serves as a high-quality, near-optimal solution for our problem. By integrating the Benders algorithm, feasibility cuts or no-good cuts are employed to exclude solution spaces that are optimal for the master problem but infeasible for the subproblem—specifically those violating battery capacity or time-window constraints—thereby rapidly identifying these high-quality solutions. Subsequently, optimality cuts are iteratively generated to further enhance solution quality and approach the global optimum. If an appropriate termination mechanism is implemented at this point, the final solution quality is guaranteed to be at least as high as—and typically superior to—the optimal V2G schedules for the best feasible routes.

(1) The master problem for ET routing optimization:

$$\text{Min } W_1 \sum_{k=1}^{K} \sum_{(i,j)\in A} c_{ij} x_{ij}^k + \varphi(x_{ij}^k) \tag{19}$$

The objective function of this model is defined by Equation (19), subject to constraints (2)-(6), (22), (23) and (24). Here, $\varphi(x_{ij}^k)$ is an auxiliary variable representing the lower bound of the subproblem model. This master problem aims to determine minimum-cost paths.

(2) The subproblem for V2G coordination scheduling:

Let $\bar{x}_{ij}^k$ be the solution to the master problem and λ_{ij}^k the dual multipliers of the corresponding constraints.

$$\varphi(\bar{x}_{ij}^k) = \text{Min} \sum_{k=1}^{K} \left(W_2 \sum_{i=1}^{N} p_i c_i^k + W_3 t_N^k + W_4 (b_N^k - E_N^{\text{set}}) \right) \tag{20}$$

$$x_{ij}^k = \bar{x}_{ij}^k(\lambda_{ij}^k), \quad \forall (i,j) \in A, \forall k \tag{21}$$

The objective function of this model is defined by Equation (20), subject to constraints (7)-(11), (15)-(18) and (21). This subproblem aims to achieve optimal V2G scheduling based on the routing solution obtained from the master problem.

(3) Cut Generation

Since the V2G coordination subproblem is a MILP model, it is first relaxed to a linear program to generate standard Benders cuts. However, to address scenarios where the linear relaxation holds but the integer constraints are violated, a specific integer cut is introduced.

Feasibility Cut: When the master solution $\bar{x}_{ij}^k$ is passed to the relaxed subproblem, if the relaxed subproblem is found to be infeasible, it indicates that the path is physically invalid even under continuous assumptions. In this case, the dual unbounded ray u_k and its intercept δ_k are retrieved to generate a feasibility cut according to Equation (22), enforcing the generation of valid routes.

No-good Cut: A special case arises when the relaxed linear subproblem is feasible, but the original MILP subproblem is infeasible due to discrete constraints (e.g., incompatible charging levels). In this scenario, standard duality-based cuts are insufficient. Instead, a simplified no-good cut is generated according to Equation (23), explicitly forbidding the current path set S_k and forcing the master problem to identify an alternative route.

Optimality Cut: When the master solution $\bar{x}_{ij}^k$ is feasible for both the relaxed linear subproblem and the original MILP, the dual multipliers λ_{ij}^k and the objective value $\varphi(\bar{x}_{ij}^k)$ are retrieved. These form an optimality cut, as defined in Equation (24), which refines the master problem's feasible region and drives the final approximate solution toward the global optimum.

$$\delta_k + \sum_{k=1}^{K} \sum_{(i,j)} u_k^\top (x_{ij}^k - \bar{x}_{ij}^k) \le 0 \tag{22}$$

$$\sum_{(i,j)\in S_k} x_{ij}^k \le |S_k| - 1, \quad \forall k \in K_{\text{inf}} \tag{23}$$

$$\varphi(\bar{x}_{ij}^k) + \sum_{k=1}^{K} \sum_{(i,j)} \lambda_{ij}^k (x_{ij}^k - \bar{x}_{ij}^k) \le \varphi(x_{ij}^k) \tag{24}$$

The upper bound of the model is given by Equation (25), while the lower bound is defined by Equation (26), where $\varphi(\bar{x}_{ij}^k)$ in the upper bound represents the objective value of the original non-relaxed subproblem.

$$\overline{\Phi} = W_1 \sum_{k=1}^{K} \sum_{(i,j)} c_{ij} \bar{x}_{ij}^k + \varphi(\bar{x}_{ij}^k) \tag{25}$$

$$\underline{\Phi} = W_1 \sum_{k=1}^{K} \sum_{(i,j)} c_{ij} x_{ij}^k + \varphi(x_{ij}^k) \tag{26}$$

During the algorithm execution, the first obtained upper bound represents the optimal solution to the two-stage distributed optimization problem—specifically, the optimal V2G scheduling scheme under the optimal feasible routing plan. This solution serves as a high-quality approximation of the global optimum. Subsequently, by iteratively introducing feasibility cuts, optimality cuts, and no-good cuts, this near-optimal solution is progressively refined toward the exact optimum. The iterative process is terminated when the improvement in the upper bound over a specified

interval falls below a predefined threshold. Additionally, computation time and the number of iterations serve as supplementary termination criteria. This ensures the efficient utilization of computational resources and the controllability of the process.

B. Graph-Based Reformulation and Algorithmic Framework

The graph-based modeling paradigm formally represents complex optimization models through graph structures, where nodes correspond to subproblems and edges capture coupling relationships between them. This structure aligns inherently with Benders decomposition, providing an ideal theoretical foundation for algorithm implementation. Following this framework, we reconstruct the Benders algorithm using Plasmo.jl [22].

First, following graph-based modeling theory and the Benders decomposition structure, we construct a graph representation of the optimization model. Since the EVRP variables and V2G coordination variables for each ET belong to different subproblems, we use $2K$ nodes to model the system: K nodes for truck routing and K nodes for V2G scheduling. Coupling constraints between and within nodes are then systematically configured according to the variable relationships in Benders decomposition and the PlasmoBender framework [23]. Specifically, independent constraints are assigned to corresponding nodes, while coupling constraints are formulated as linking constraints between nodes.

The resulting graph topology is shown in Fig. 3, which clearly reveals inherent variable relationships. If each ET's variables are represented by gray nodes, the Benders decomposition partitions $2K$ nodes into two distinct sets: The truck routing problem–formulated as the master problem–is represented by green nodes in the root graph, while the V2G scheduling subproblems are denoted by red nodes in the subgraphs. For the nodes corresponding to ET k, the objective function of the root graph nodes is defined as $\text{Min} \sum_{(i,j)\in A} c_{ij} x_{ij}^k$, subject to constraints (2), (4) and (5). The objective function of subgraph nodes k is defined as $\text{Min}\left(W_2 \sum_{i=1}^{N} p_i c_i^k + W_3 t_N^k + W_4 (b_N^k - E_N^{\text{set}})\right)$, subject to constraints (7)-(11) and (15)-(18) . Additionally, the linking constraints between subgraph nodes k and root graph nodes k are given by Equations (21), while the linking constraint among all ET root graph nodes is provided by Equation (3).Furthermore, for homogeneous fleets, the linking constraint between adjacent root graph nodes is given by Equation (6).

Following the pseudocode outlined in Algorithm 1, the proposed Graph-Benders decomposition algorithm efficiently yields a solution whose quality is no less than that of the optimal V2G coordination scheme under the optimal feasible routing solution.

V. Case Study

Three experiments are conducted to validate the proposed algorithm. First, transportation network tests involving multiple representative networks verify the model's adaptability. Second, split-delivery scenarios—based on a network from the first experiment and incorporating localized nodes with heavy cargo loads—demonstrate the feasibility of split delivery and the algorithm's performance under extreme distributions. Finally, two sets of large-scale test cases are used to confirm computational efficiency.

Light-duty ETs are mainly deployed for short-haul distribution, whereas medium- and heavy-duty ETs undertake long-distance cross-provincial trunk transportation with multi-stop nodes, featuring larger load capacities and superior charging/discharging performance. Their

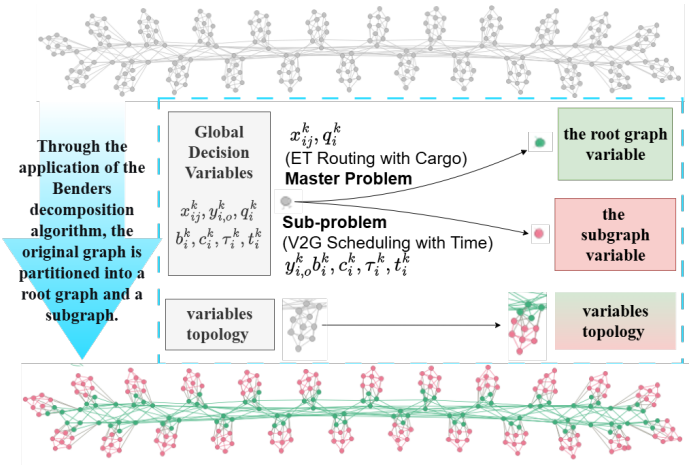


Fig. 3: Graph Model Topology for ET Routing and V2G Scheduling

Algorithm 1 Graph-Benders Decomposition Algorithm

Input: Master graph model framework and related data
Output: Current optimal path and scheduling scheme

```
Initialize upper bound, set detection step size and threshold
loop
    Solve master graph model for routing plan
    Solve subgraph model (MILP) for V2G coordination
    if MILP Subgraph Model is infeasible then
        Solve linear relaxation of subproblem
        if linear relaxation is infeasible then
            Construct feasibility cut (Equation (22)),
            add to master graph model
        else
            Generate no-good Cut (Equation (23))
            add to master graph model
        end if
    else
        Update global optimal upper bound and solution
        Calculate the upper bound variation for the set step
        if variation > threshold then
            Solve linear relaxation of subproblem
            Construct optimality cut (Equation (24))
            add to master graph model
        else
            Output optimal path and scheduling scheme;
            Break loop,algorithm terminate
        end if
    end if
end loop
```

complicated operational scenarios fit the proposed optimization model well. Considering the prevalent application of medium-duty ETs in the logistics industry and the expandability of the developed model toward heavy-duty ET applications, medium-duty ETs are selected as the research subject.

Following the operational data-driven classification for ETs [5], medium-duty trucks are defined as vehicles with a Gross Vehicle Weight (GVW) between 4.5 and 12 tonnes, which typically corresponds to a payload range of approximately 2–6 tonnes. To maintain consistency with this industry standard, the fleet in this study is categorized into three heterogeneous types of medium-duty ETs. Based on commercial specifications from the BYD T-series and Isuzu Series, these are designated as: ET-I (3,000 kg), ET-II (4,000 kg), and ET-III (5,000 kg), with corresponding battery capacities of 100 kWh, 140 kWh, and 180 kWh, respectively. For all vehicle types, the state-of-charge is strictly maintained between 10% and 90% of the maximum capacity. Additional operational parameters include a consistent travel speed of 1.5 km/min, energy consumption of 0.4 kWh/km, and an operating cost of ¥0.34/km. To support these operations, the model configures two types of charging nodes with distinct power-tariff profiles: high-power stations (288 kW at ¥1.8/kWh) and standard-power stations (120 kW at ¥1.0/kWh) [27].To standardize V2G interactions, the discharge power limit is capped at 120 kW for both node types . Finally, objective weights are set as $W_1 = W_2 = 1$, $W_3 = 0.01$, and $W_4 = 1.4$, with initial and terminal energy levels set to $E_0^{\text{set}} = 0.9E_k^{\max}$ and $E_N^{\text{set}} = 0.5E_k^{\max}$.

Three methods are compared:

- M1: Direct solution via Gurobi (benchmark)
- M2: Iterative LP decomposition from [20], which alternates between routing and charging subproblems
- M3: The proposed Graph-Benders algorithm

All experiments run on an Intel i5-10200H processor with 16 GB RAM, using Julia v1.10.4, JuMP v1.29.3, Gurobi v11.0.2, Plasmo v0.7.1, and PlasmoBenders v0.3.1.

A. Diverse Transportation Networks Tests

This study utilizes three representative transportation networks—Chengdu-Chongqing (Test 1), Shijiazhuang-Beijing (Test 2), and Shijiazhuang-Jinan (Test 3)—to conduct comprehensive evaluations across diverse geographical and operational environments. These geographically diverse networks span different regions of China with varying traffic patterns and demand characteristics, providing comprehensive testing scenarios for validating the model's generalization capability. Prior to experiments, Dijkstra's algorithm preprocesses each network by computing shortest paths between service nodes, simplifying the infrastructure into graph $G(N, A)$ to establish a standardized foundation for evaluating the graph-based model and Graph-Benders algorithm's effectiveness. Experimental Parameters as follows:

- Transportation networks: The three test networks are shown in Fig. 4.
- Service nodes: Charging station specifications and cargo weights at each service node (Table I) remain constant during operation. Service request times are estimated based on inter-city distances and historical traffic data via simulation modeling.
- ET types: One ET-I, one ET-II, and one ET-III are deployed.

TABLE I: Service Node Specifications

Node 1-9				Node 10-17			
Node	P_i (¥)	$P_{ch,i}$ (kW)	Q_i (kg)	Node	P_i (¥)	$P_{ch,i}$ (kW)	Q_i (kg)
1	–	–	2000	10	1.0	120	400
2	1.0	120	700	11	1.8	288	600
3	1.8	288	500	12	1.0	120	500
4	1.0	120	600	13	1.8	288	500
5	1.8	288	800	14	1.0	120	300
6	1.0	120	1500	15	1.8	288	600
7	1.8	288	800	16	1.0	120	1200
8	1.0	120	500	17	–	–	0
9	1.8	288	500				

TABLE II: Computational Speed Comparison in Test1-3

Test	M1 (Gurobi)		M2 (ILP)	M3 (Graph-Benders)
Test 1	176s/747.96	315s/747.96	4.3s/762.14	1.83s/747.96
Test 2	214s/812.20	532s/812.20	1.24s/815.68	0.17s/812.20
Test 3	94s/667.36	3000s/667.36	0.46s/683.47	0.14s/667.36

Table II details the performance metrics of the three methods. Using M1 (Gurobi) as a baseline, the table records the time required to obtain the first feasible solution superior or equivalent to that of M3, as well as the time to reach final convergence (or the solution obtained when the computation time exceeds 3,000 s) . For the heuristic algorithms M2 and the proposed M3 (Graph-Benders), the final objective values and the initial computation times are reported. The same methodology applies to Table IV and Table VI in the subsequent experiments.

The results,as illustrated in Table II, indicate that while M1 theoretically guarantees global optimality, it incurs a substantial computational burden. Specifically, M1 requires a considerable amount of time even to find a feasible solution of comparable quality to M3.Furthermore, M1's computational efficiency exhibits high sensitivity to data characteristics, leading to significant performance fluctuations across different instances; for example, in Test 3, Gurobi failed to converge even within the 3,000s limit. In contrast, the heuristic approaches M2 and M3 demonstrate an order-of-magnitude improvement in efficiency, achieving sub-second solution times.Crucially, M3 identifies superior feasible solutions more rapidly than M2. These solutions are strictly consistent with M1's global optima or the best feasible solutions reached by M1 at the 3,000 s limit, demonstrating that M3 achieves the optimal balance between computational efficiency and solution quality.

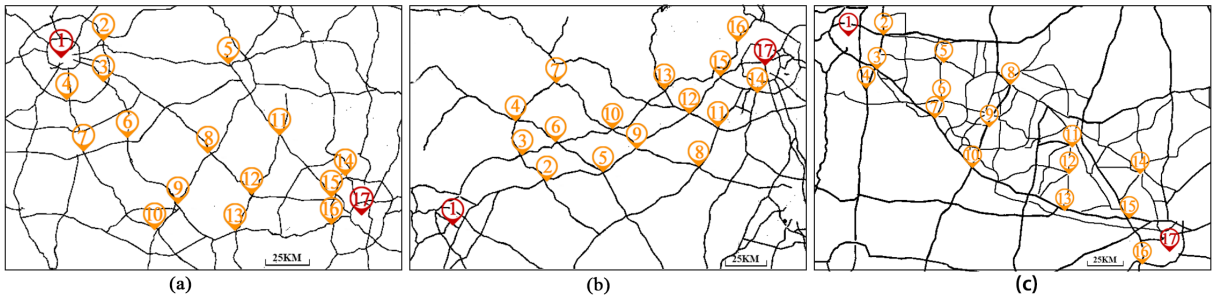


Fig. 4: Validation on Multi-Region Transportation Networks
: (a) Chongqing to Chengdu network, (b) Shijiazhuang to Beijing network, (c) Shijiazhuang to Jinan network

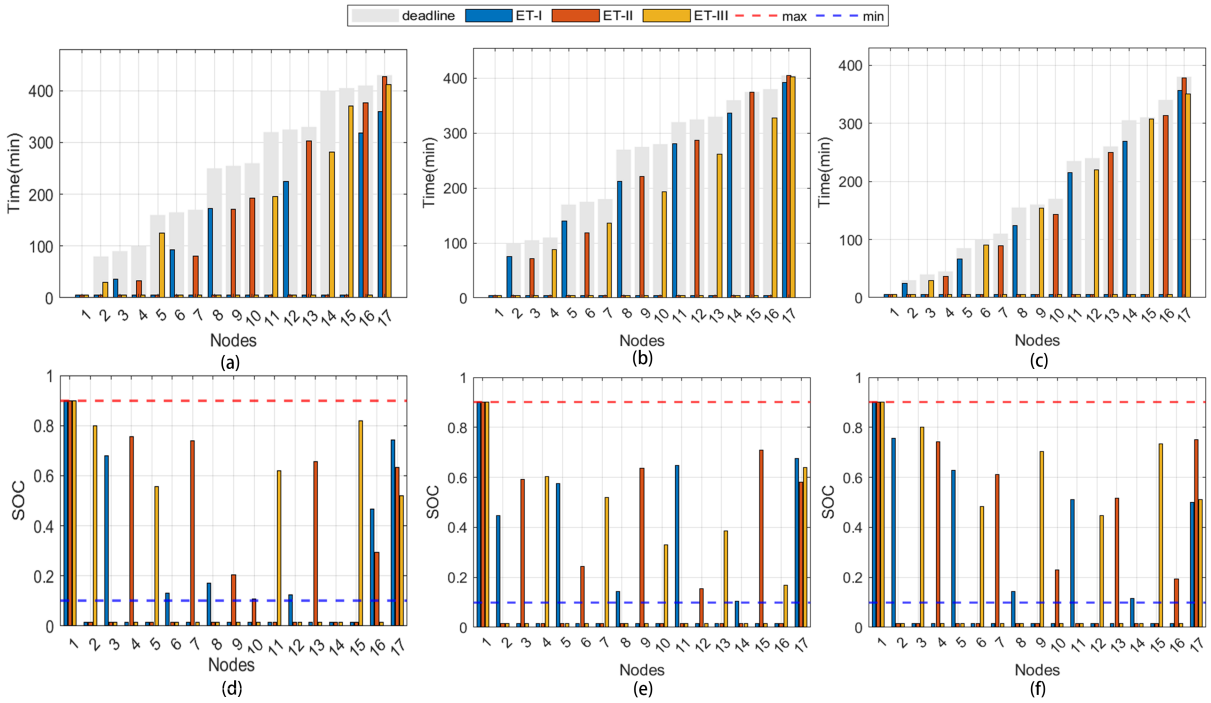


Fig. 5: Time and SOC Profiles Across Test Networks: (a)-(c) arrival time by truck type for Tests 1-3; (d)-(f) SOC at nodes by truck type for Tests 1-3

Based on the Graph-Benders results from each experimental set, the performance data are visualized in Fig. 5. The temporal profiles demonstrate that all ETs strictly satisfy the time window constraints at their respective nodes while revealing the node visitation sequences. Meanwhile, the SOC profiles show that the energy consumption remains within the prescribed bounds of 10%–90% throughout the journeys, with terminal SOC levels consistently exceeding 50%. These outcomes collectively validate the temporal and energetic feasibility of the solutions obtained by the Graph-Benders algorithm, as well as its generalizability across different transportation networks. The following subsection further examines the model's capability in handling split-delivery requirements at service nodes.

B. Split-Delivery Capability Tests

This section examines the model's split-delivery capability under specialized cargo weight distributions—where cargo loading is concentrated at certain nodes and requires splitting to be loaded onto multiple ETs. The model's split-delivery capability effectively handles imbalanced cargo weight distributions by enabling flexible load allocation, which mitigates routing infeasibility caused by battery capacity or time window constraints and enhances computational tractability. The experimental setup evaluates the algorithm's adaptability through the following configuration:

Using the Chengdu-Chongqing (Test 1)transportation network—Test4 and 5 are variants of Test 1 featuring overloading concentrated at different nodes—we adjust the cargo weights at service nodes as specified in Table III. The table exhibits distinct characteristics of severe localized

heavy loading, which precisely aligns with the core setting of split-delivery capability test : for example, Node 10 in Test 5 carries a heavy load of 4000 kg and Node 6 in Test 4 reaches 3500 kg far exceeding the weight of other nodes.

TABLE III: Cargo Distribution for Tests 4-5 (kg)

Nodes 1-9				Nodes 10-17			
Node	Test 1	Test 4	Test 5	Node	Test 1	Test 4	Test 5
1	2000	200	200	10	400	400	4000
2	700	200	200	11	600	500	200
3	500	400	700	12	500	800	600
4	600	600	600	13	500	900	800
5	800	400	400	14	300	900	800
6	1500	3500	200	15	600	400	1000
7	800	1000	500	16	1200	700	700
8	500	800	300	17	0	0	0
9	500	300	800				

Table IV details the computational performance metrics in split-delivery scenarios. Notably, in this set of experiments, the severe imbalance in cargo weight distribution significantly constricts the feasible region, thereby increasing the difficulty of finding feasible solutions. Consequently, the time required for M1 to obtain a solution superior or equivalent to the quality of M3 is noticeably delayed. For M3, the increased complexity may necessitate more Benders cuts to reach feasibility, leading to a moderate increase in its computation time compared to the network tests.In contrast, M3 does not suffer from a significant degradation in performance. It not only surpasses M2 in computational speed but also remains strictly consistent with M1's global optimal objective values.

TABLE IV: Computation Performance in Test 4-5

Test	M1 (Gurobi)		M2 (ILP)	M3 (Graph-Benders)
Test1	176s/747.96	315s/747.96	4.3s/762.14	1.83s/747.96
Test 4	199s/797.13	258s/797.13	12.82s/803.74	0.367s/797.13
Test 5	416s/845.61	438s/845.61	16.79s/868.54	1.04s/845.61

Fig. 6 presents the cargo load distribution obtained from M3, showing both the demand at each logistics node and the load assigned to individual ETs. In scenarios with concentrated demand, multiple trucks collaboratively serve the same node, as observed at Node 8 in Test 4 and Node 10 in Test 5 . Crucially, the cargo load of any vehicle remains within the node's total demand (represented by the length of the shaded bar). This result not only validates the model's split-delivery capability but also fully demonstrates the efficiency and applicability of the M3 algorithm.

C. Large-Scale Test

Large-scale tests are primarily designed to evaluate the computational performance of the proposed algorithm on high-dimensional instances. The experimental setup comprises two distinct groups. First, the problem scale is expanded in Test 6 by increasing the number of service nodes and ETs within the Chengdu-Chongqing transportation network. Second, a long-distance network spanning from Zhengzhou to Beijing is introduced as Test 7 to further assess the model's scalability.

For Test 6, the number of service request nodes is increased to 32, with their spatial distribution illustrated in Fig. 7. The fleet consists of six vehicles, comprising two units each for the three categories of ETs. The detailed cargo load distribution across nodes is provided in Table V, totaling a weight of 24,000 kg. Under this high-complexity configuration, the resulting optimization model encompasses 19,585 decision variables.

TABLE V: Cargo Distribution for Test6

Node ID	Weight Distribution (kg)
1–8	2500, 700, 500, 1600, 800, 500, 800, 500
9–16	1200, 700, 600, 500, 500, 1300, 600, 200
17–24	700, 1100, 600, 800, 500, 1300, 500, 500
25–32	900, 600, 500, 400, 300, 1600, 200, 0

TABLE VI: Computational Efficiency in Test6

Test	M1 (Gurobi)		M2 (ILP)	M3 (Graph-Benders)
Test 6	–/–	10000s/1516	438s/1601	27.73s/1512

The experimental results for Test 6, presented in Table VI, demonstrate that M3 significantly outperforms the other solution approaches in terms of computational efficiency. Specifically, M3 attains a near-optimal solution with an objective value of 1512 in only 27.73 seconds. In contrast, even after reaching the maximum time limit of 10,000 seconds, the direct method (M1) merely obtains a feasible solution with an objective value of 1516 and fails to converge. Additionally, the M2 algorithm requires 438 seconds only to yield a feasible solution of 1601. In summary, Graph-Benders exhibits superior performance in both solution speed and quality.

Due to the substantial growth in data volume and node scale in Test 6, coupled with the increased number of ETs, energy interactions within the system have become increasingly frequent. Accordingly, Figure 8 illustrates the energy management scheme generated by M3, quantitatively characterizing the interaction dynamics of six ETs across 32 nodes. In this visualization, red and blue regions denote charging and discharging behaviors, respectively—exemplified by ET-III 2 discharging 90 kWh at Node 13 and charging 130 kWh at Node 18. These results intuitively validate the deep energy interaction mechanism between the electric logistics fleet and the grid within complex networks.

For Test 7, the problem scale is further expanded to 64 service request nodes, as depicted in Fig. 9, with a total cargo load of 48,000 kg. The fleet size is increased to 12 vehicles, featuring four units per ET category. Due to the substantial increase in transport distance, the battery capacity of all ETs is standardized to 300 kWh for simulation feasibility, with the load capacity of each ET type kept unchanged. Under this ultra-high-complexity configuration, the generated optimization model contains 127489 decision variables.

Due to the excessive scale of the problem, M1 could not identify a feasible solution within 20,000s. However, M3

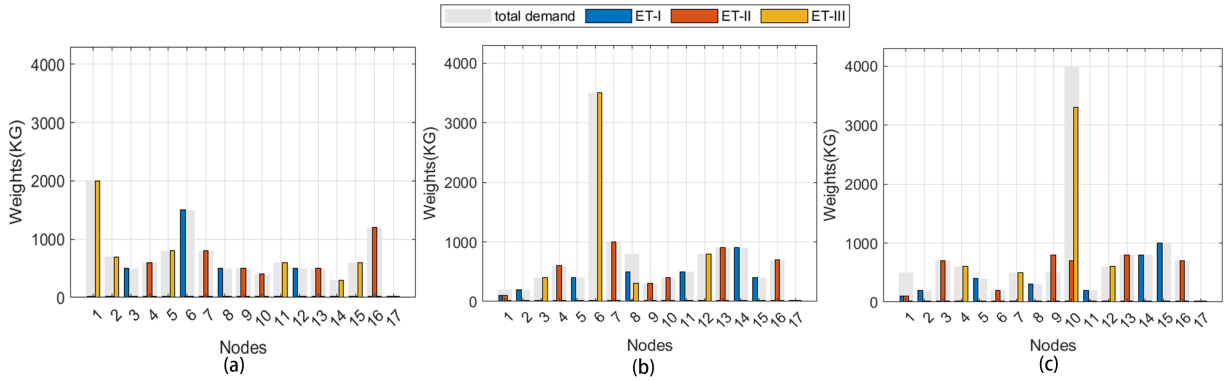


Fig. 6: Cargo Load Distribution in Split-Delivery Tests
: (a) Test 1, (b) Test 4, and (c) Test 5 scenarios, showing loads allocated to each truck type and total nodal demand.

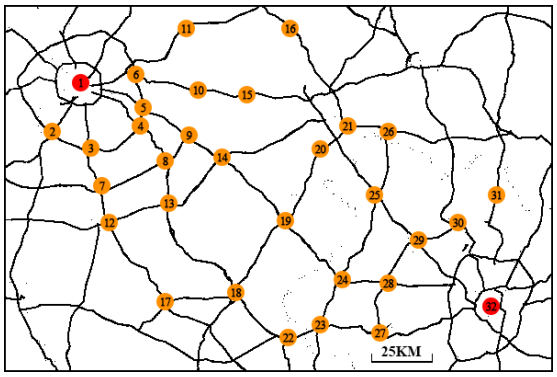


Fig. 7: 32-Node Layout of the Chengdu–Chongqing Transportation Network

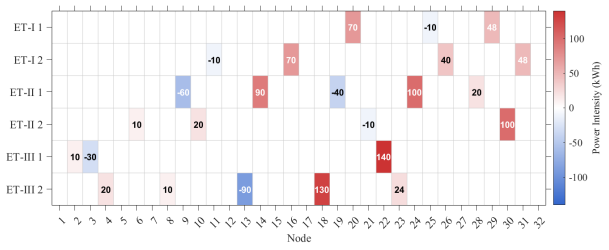


Fig. 8: V2G Energy Interaction Spatiotemporal Distribution in Test 6.

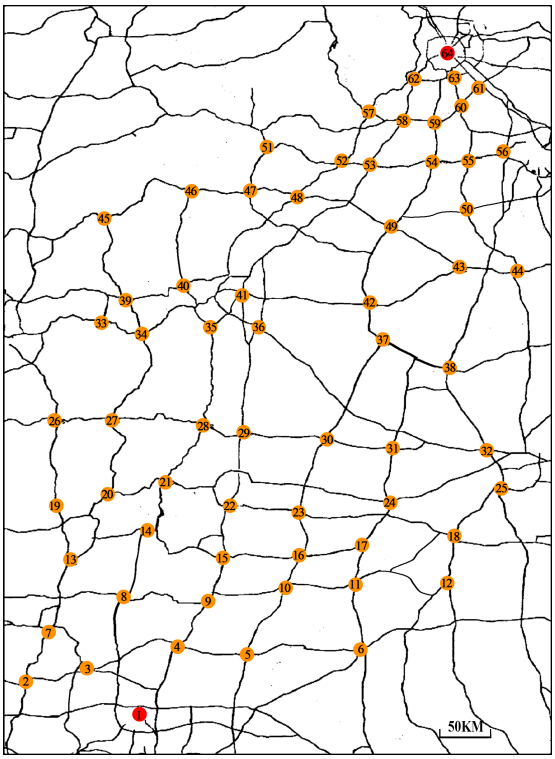


Fig. 9: 64-Node Layout of the Zhengzhou–Beijing Transportation Network

yielded a solution of 10,020 in just 51s, with the objective value further decreasing to 9,334 by 130s, as illustrated in Figure 7. The reduction of the upper bound is primarily driven by the continuous generation of optimality cuts, which play a pivotal role in guiding the master problem to search for superior feasible solutions and approaching the global optimum.

In summary, the large-scale network experiments validate the superior performance of the Graph-Benders algorithm in addressing high-dimensional and complex problems.

VI. Conclusion

This study focuses on addressing the core challenge of tripartite optimization—routing, charging, and cargo loading—for ETs in the context of V2G integration under stringent time constraints, a problem that is NP-hard in nature. Recognizing that transportation costs represent a large proportion of total costs and that minimum-cost paths offer time redundancy to optimize V2G costs, this paper proposes the Graph-Benders algorithm, based on an exact logic-based Benders approach, which integrates graph-based modeling with Benders decomposition. By generating feasibility cuts and no-good cuts, the algorithm efficiently prunes infeasible candidate solutions to achieve

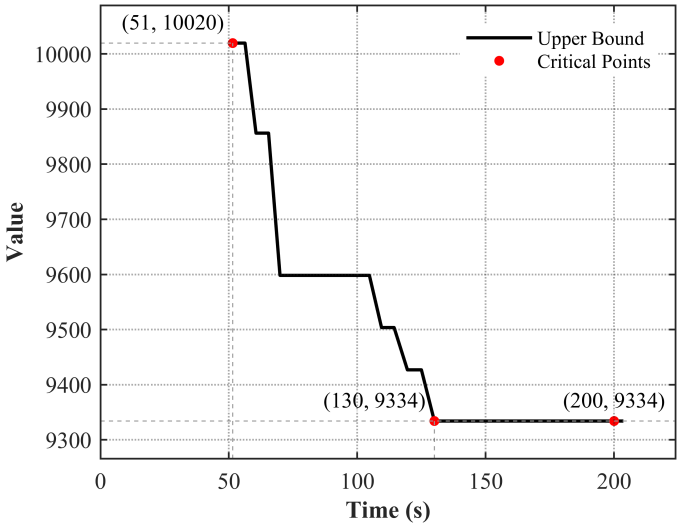


Fig. 10: Variation of Upper Bound for Test 7

optimal V2G scheduling while rapidly identifying optimal routes. Furthermore, the inclusion of optimality cuts further enhances solution quality, ensuring that the algorithm yields high-quality solutions that match or exceed the performance of the optimal V2G schedule for the best feasible routes

Experimental validation confirms the proposed method's superiority. Transportation network tests across diverse geographic scenarios, combined with split-delivery experiments, compellingly demonstrate the proposed algorithm's robust adaptability in EVRP and V2G optimization. Crucially, it achieves significant efficiency gains while matching the solution quality of Gurobi. In large-scale experiments with 19,585 decision variables, the proposed Graph-Benders decomposition requires only 28 seconds to obtain a solution superior to the one achieved by Gurobi under the maximum time limit of 10,000 seconds. At a scale of 127,489 variables—where Gurobi fails to yield a solution within 20,000 seconds—the algorithm identifies a high-quality near-optimal solution in only 130 seconds. Consequently, the method is well-suited for the real-time scheduling of large-scale ET logistics. By effectively addressing the efficiency and robustness requirements of ET operations, this algorithm provides critical technical support for the widespread adoption of ETs, thereby accelerating the transition toward green and low-carbon supply chains.

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